Task	Prompt	Answer
dominating set	The task is to determine the dominating set of a graph. A dominating set is a subset of nodes such that every node in the graph is either in the set or adjacent to a node in the set. For directed graphs, any node not in the dominating set must be a successor of a node within the set. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 2), (1, 5), (1, 6), (1, 7), (2, 3), (2, 4), (5, 6), (7, 3), (7, 4). Question: What is the dominating set of the graph? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[1, 3, 4]
edge existence	The task is to determine if there is an edge connecting two nodes. For an undirected graph, determine if there is an edge between nodes *u* and *v*. For a directed graph, determine if there is an edge from *u* to *v*. Here is an undirected graph containing nodes from 1 to 8. The edges are: (1, 2), (1, 6), (3, 8), (3, 4), (8, 4), (8, 5), (8, 7), (4, 7), (4, 5), (7, 5). Question: Is there an edge between node 5 and node 3? Your answer should be Yes or No.	No
edge number	The task is to determine the number of edges in the graph. For the undirected graph, you should count the edge between two nodes only once. Here is an undirected graph containing nodes from 1 to 10. The edges are: (1, 10), (1, 8), (10, 7), (8, 6), (2, 5), (2, 4), (2, 6), (5, 4), (5, 9), (4, 3), (4, 9), (3, 7). Question: How many edges are there in the graph? Your answer should be an integer.	12
global efficiency	The task is to determine the global efficiency of a graph. Global efficiency is the average efficiency of all pairs of nodes. The efficiency of a pair of nodes is the multiplicative inverse of the shortest path distance between the nodes. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 5), (1, 4), (5, 2), (2, 7), (7, 3), (3, 6). Question: What is the global efficiency of the graph? You need to format your answer as a float number.	0.5310
hamiltonian path	The task is to return a Hamiltonian path in a directed graph. A Hamiltonian path is a path in a directed graph that visits each vertex exactly once. The input graph is guaranteed to be directed and tournable. Here is a directed graph containing nodes from 1 to 8. The edges are: (2, 1), (2, 4), (2, 5), (2, 6), (2, 7), (1, 3), (1, 4), (1, 7), (3, 2), (3, 7), (3, 8), (4, 3), (4, 5), (4, 7), (5, 1), (5, 3), (5, 8), (6, 1), (6, 3), (6, 4), (6, 5), (7, 5), (7, 6), (8, 1), (8, 2), (8, 4), (8, 6), (8, 7). Question: Return a Hamiltonian path in the graph. You need to format your answer as a list of nodes, e.g., [node-1, node-2, ..., node-n].	[2, 1, 4, 5, 3, 8, 7, 6]